MULTILSB Library

Kerem Akartunalı * Andrew J. Miller †

Revised: April 2010

1 Formulation

MULTILSB (Multi-Item Lot-Sizing with Backlogging) library considers the lot-sizing problems with the following basic formulation:

$$\min \sum_{t=1}^{NT}\sum_{i=1}^{NI} bc^i b_t^i + \sum_{t=1}^{NT}\sum_{i=1}^{NI} hc^i s_t^i \tag{1}$$

$$\text{s.t. } x_t^i + s_{t-1}^i - s_t^i + b_t^i - s_{t-1}^i = d_t^i \qquad t \in [1,NT], i \in endp \tag{2}$$

$$x_t^i + s_{t-1}^i - s_t^i = \sum_{j \in \delta(i)} r^{ij} x_t^j \qquad t \in [1,NT], i \in \overline{endp} \tag{3}$$

$$x_t^i \leq d_{1,NT}^i y_t^i + b_{t-1}^i \qquad t \in [1,NT], i \in endp \tag{4}$$

$$x_t^i \leq r^{ij}\Big(\sum_{j \in endp} d_{1,NT}^j y_t^i + b_{t-1}^i\Big) \qquad t \in [1,NT], i \in \overline{endp} \tag{5}$$

$$y_t^i \leq w_t^f \qquad t \in [1,NT], i \in [1,NI], f \in [1,NF] \tag{6}$$

$$\sum_{i=1}^{NI} a_k^i x_t^i + \sum_{f=1}^{NF} ST_k^f w_t^f \leq C_t^k \qquad t \in [1,NT], k \in [1,NK] \tag{7}$$

$$a_k^i x_t^i \leq (C_t^k - ST_k^f) y_t^i \qquad t \in [1,NT], i \in [1,NI], k \in [1,NK], f \in [1,NF] \tag{8}$$

$$y \in \{0,1\}^{NT \times NI}, w \in \{0,1\}^{NT \times NF}, \quad x, s, b \in R^{NTxNI} \tag{9}$$

*Dept. of Management Science, University of Strathclyde, Glasgow G1 1QE, UK. E-mail: Kerem.Akartunali@strath.ac.uk

†IMB, Université de Bordeaux 1; RealOpt, INRIA Bordeaux Sud-Ouest, France. E-mail: Andrew.Miller@math.u-bordeaux1.fr

Sets:

NT: Number of time periods in the horizon

NI: Number of items

NK: Number of machines

NF: Number of product families (each item belongs to one family)

$endp$: The set of all end-items

$\delta(i)$: The set of all successor items of item i

$\overline{endp}$: The set of all items other than end-items, i.e. $\overline{endp} = [1, NI]/endp$

Variables:

x_t^i: Production amount in period t of item i

s_t^i: Inventory carried at the end of period t to period $t+1$ of item i

b_t^i: Backorder amount to the period t from period $t+1$ of item i

y_t^i: 1 if production of item i occurs in period t, 0 otherwise

w_t^f: Setup variable for each family and period; 1 if production of any item in the family of f occurs in period t, 0 otherwise

Parameters:

bc^i: Backlogging cost for each item i per period

hc^i: Holding cost for each item i per period

d_t^i: Demand for end-item i in period t

$d_{1,NT}^i$: Total demand for end-item i through horizon

r^{ij}: Amount of item i required to produce one unit of item j

a_k^i: Variable production time per unit of item i on machine k

ST_k^f: Setup time of product family f on machine k

C_t^k: Production capacity on machine k in period t

Constraints:

(1): Objective function includes only inventory holding and backlogging costs, but no setup or production costs.

(2): Flow balance for end-items

(3): Flow balance for non-end-items

(4): Relation between the linear production and binary variables for end-items

(5): Relation between the linear production and binary variables for non-end-items

(6): Relation between two different binary variables

(7): Capacity constraint for each machine and period

(8): Another relation between the linear production and binary variables

(9): Integrality and nonnegativity requirements

2 More Details for Data Generation

The problem structure and data generation mainly is derived from the MIFO (Multiple Item, Family-Oriented Fixed Costs) experiment set of Simpson and Erenguc [3]. The set considers problems with multiple items, multiple production stages and the items are grouped into product families due to similarities, as seen in the formulation in the previous chapter.

There are 4 sets of test problems, and each set consists of 30 instances. Except SET2, which has 24 periods, all the problems have 16 periods. All the problems have 78 items, of which only 6 are end-items, i.e., these have external demand. Each item belongs to a specific product family, and there are 11 families total. There are 6 machines, and 3 of these machines only consider setup times, in addition to variable processing times.

The BOM structure, variable production times and echelon holding costs are the same as the data of Simpson and Erenguc [3]. The six "families" (namely 1,2,8,9,10,17) of the same paper are considered as machines in our test problems. The backlogging costs are set to be double of inventory holding cost in SET1 and SET2, and to be 10-times of inventory holding cost in SET3 and SET4. The resource utilization factor is set to be higher than total demand in all the sets (namely 1.05 in SET1 and SET2, 2 in SET3, and 1.25 in SET4), forcing the test problems allow backlogging to the last period from beyond the horizon, which makes the decisions harder. The demand data is generated in a similar fashion to Simpson and Erenguc [3], where the normal distribution is used to generate demands and then all negative values are set to zero. The details for means and standard deviations (SD) used can be seen in Table 1. Also note that the same demand data is used over SET1, SET3 and SET4, i.e., the same instance number in all these three test sets have exactly the same demand data, and the only difference among these sets is then that they do have different resource utilizations and backlogging costs. Demand data is not extensively provided in this document, however, interested reader should see the individual demand files in the .zip file for each different instance.

End-items	Variability	Mean	SD	Replications
1-4	Low	100	10	10
	Medium	100	50	10
	High	100	90	10
5-6	Low	40	4	10
	Medium	40	20	10
	High	40	35	10

Table 1: Demand Data

In our previous studies, we used these instances for extensive computational tests. We refer the interested reader to [1] for results comparing upper bounds obtained mainly using IP heuristics, and to [2] for detailed reporting of lower bound results with a variety of different techniques.

References

[1] K. Akartunalı and A.J. Miller. A heuristic approach for big bucket multi-level production planning problems. *European Journal of Operational Research*, 193:396–411, 2009.

[2] K. Akartunalı and A.J. Miller. A computational analysis of lower bounds for big bucket production planning problems. Under Revision, Available at *Optimization Online*, `http://www.optimization-online.org/DB_HTML/2007/05/1668.html`, 2010.

[3] N.C. Simpson and S.S. Erenguc. Modeling multiple stage manufacturing systems with generalized costs and capacity issues. *Naval Research Logistics*, 52:560–570, 2005.